LIU XINGLE

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Collaborative Double Major Programme | Incoming Year 3 | ASEI + ECE

EDUCATION

- **The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) & The Chinese University of Hong Kong (CUHK)** *Sep. 2024 - Present*
  Collaborative Double Major Programme; Undergraduate Hong Kong & Shenzhen, China
  - Majors: Aerospace Science and Earth Informatics; Electronic and Computer Engineering
  - CGPA: **3.829**/4.00 (**7**/162 in ECE)
  - MGPA: **3.889**/4.00
  - 2025–2026 YGPA Rank: **1**/160 in ECE

RESEARCH INTERESTS

Embodied AI; **Vision-Language Navigation (VLN)**; LLM Compression; Reinforcement Learning & Imitation Learning; **Nighttime-Light Remote Sensing**

RESEARCH EXPERIENCE

- **Control and Decision Lab, CUHK-Shenzhen** *Sep. 2025 - Present*
  Research Assistant | Supervisors: Prof. Xiaoqiang Ji & Dr. Zike Yuan Shenzhen, China
  Keywords: Vision-language navigation, embodied AI, LLM, RL & IL, diffusion policy, edge deployment.
  - Designed a **lightweight LLM-based vision-language navigation agent** for long-horizon navigation and dynamic obstacle avoidance, trained through knowledge distillation from a reinforcement-learning-based obstacle-avoidance expert and supervised fine-tuning on an expert navigation dataset.
  - Integrated a diffusion-based action head and deployed the model on UAV edge hardware using quantization-based compression, enabling **few-shot generalization and zero-shot navigation**.
- **Department of Geography and Resource Management, CUHK** *Mar. 2026 - Present*
  Undergraduate Researcher | Supervisor: Prof. Qiming Zheng Hong Kong, China
  Keywords: Remote sensing, urban afforestation, nighttime-light classification.
  - Investigated future **urban afforestation potential across the Greater Bay Area** using Sentinel-2 imagery, CCDC-based land-cover mapping, and PLUS land-use simulation; examined trade-offs between carbon sequestration and equitable tree exposure to support spatially targeted nature-based solutions.
  - Conducted **nighttime-light remote sensing** research focused on classification and applications in urban monitoring and change detection.
- **ACM Turing Award Celebration Conference - China 2025 (ACM TURC 2025)** *Oct. 2025*
  Attendee Shenzhen, China
  - Engaged with Turing Award laureates and leading scholars on advances in generative AI, large language models, and AGI.

PUBLICATION

- Hanyue Luo, Yifei Zhao, Xia Liao, **Xingle Liu**, Qiming Zheng, and Harry F. Lee (2026). **Strategic urban afforestation in the Greater Bay Area towards 2050: Spatiotemporally explicit assessment of potential and trade-offs**. *Urban Forestry & Urban Greening*, 123, 129553.

SKILLS

- **Research Skills:** Vibe Coding
- **Programming Languages:** Python, Java, MATLAB
- **Specialized Area:** LLaMa Factory, StarVLA, ArcGIS Pro, Google Earth Engine

Honors and Awards

- **National Scholarship** *2025–2026*
  Ministry of Education of the People's Republic of China
  ◦ China's highest national scholarship for undergraduates, recognizing academic excellence and outstanding all-round achievement.
- **Undergraduate Research Award (31st Round)** *Jul. 2026 - Present*
  The Chinese University of Hong Kong, Shenzhen
  ◦ Selected for an undergraduate research award with a monthly stipend of RMB 1,000.
- **Dean's List** *2024-2025*
  School of Science and Engineering, The Chinese University of Hong Kong, Shenzhen
  ◦ Outstanding academic performance during the 2024-25 academic year
- **Provincial Second Prize (Guangdong) in CUMCM** *Oct. 2025*
  China Undergraduate Mathematical Contest in Modeling Organizing Committee
  ◦ Won the award in China's largest national mathematical modeling contest (CUMCM), ranking in the top tier of the highly competitive Guangdong region.
- **Successful Participant, Mathematical Contest in Modeling (MCM)** *May. 2026*
  Consortium for Mathematics and Its Applications (COMAP)
  ◦ Recognized as a Successful Participant as part of a three-member team.
- **Second Prize, SME MC Business Case Competition** *Nov. 2024*
  School of Management and Economics, CUHK-Shenzhen
  ◦ Led a four-person team in market-entry strategy development, financial analysis, and final pitching.

Leadership and Volunteering

- **Volunteer Teacher (Astronomy) | Summer Teaching Programme** *May 2026 - Present*
  Meitan Zhejiang University Primary School, Zunyi
  ◦ Collaborating on programme preparation and developing astronomy-themed lessons for delivery to primary school students during the volunteer teaching programme in August 2026.
- **Executive Committee Member | CDMP Society** *Sep 2025 - Present*
  Collaborative Double Major Programme Society, CUHK(SZ)
  ◦ Planned and delivered a diverse range of student activities, managed materials procurement, and supported peers as a core member of the society's management team.
- **Team Presenter | 2nd "Shenxian Lake" International Innovation & Entrepreneurship Competition** *Dec. 2024*
  The Chinese University of Hong Kong, Shenzhen
  ◦ As the main speaker, participated in the collaborative creation of pitch slides and a business model canvas for a **reusable electric pump rocket project**, and presented the startup proposal to investors and reviewers.
- **Program Designer and Student Assistant | Office of Student Affairs (OSA)** *Sep. 2024 - Present*
  The Chinese University of Hong Kong, Shenzhen
  ◦ Planned and delivered multiple campus music events, coordinating logistics, performers, and on-site operations as a core OSA project-team member.